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Satellite data reveal how Sudd wetland dynamics are linked with globally-significant methane emissions

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E-mail: ajh13@aber.ac.uk**Keywords:** Sudd wetlands, wetland dynamics, satellite imaging, methane emissions, Landsat, Google Earth EngineSupplementary material for this article is available [online](#)

Abstract

Recent work has highlighted the large role of methane emissions from the Sudd wetland and surrounding ecosystems on the global atmospheric growth rate of methane since 2010. These emissions are driven by high rainfall over basin catchments linked with the positive phase of the Indian Ocean Dipole. We reconstruct flood inundation for the Sudd wetland over a 38-year period at a spatial resolution of 30 m using a new satellite Earth Observation (EO) wetland mapping tool. We reveal considerable changes in the wet season extent of the wetland, including an increase >300% since 2019 compared to the median 1984–2022 extent. We report major increases in flood extent within grassland-dominated floodplains outside of the area currently defined Sudd wetland region. These year-to-year changes in wetland extent are corroborated with total water storage anomalies inferred from satellite data (Pearson correlation $R = 0.92$), Lake Victoria levels ($R = 0.73$), and anomalies in reported annual mean global methane growth rates since 2009 ($R = 0.88$). Our analysis shows that flood water inundation is dominated by inundated vegetation and aquatic vegetation, accounting for an average of 40% and 50% of total extent, respectively, compared to open water that accounted for just 9% of inundation in a typical year. This is consistent with recent studies that report wetland methane emissions are focused on areas with inundated vegetation. Our findings also support recent studies that highlight the significant role of the Sudd wetland in driving anomalously large global atmospheric annual growth rates, 2020–2022. By capturing high resolution information on inundated vegetation, our EO wetland mapping tool has significant potential for improved wetland emission estimates of methane. Vascular plants common in the Sudd wetland, e.g. macrophytes including *Phragmites Australis* and *Cyperus Papyrus*, seem to play a key role in methane emissions and we recommend they should be the focus of future research.

1. Introduction

A wetland is broadly defined as an ecosystem that is flooded or saturated permanently or seasonally, typically by groundwater, that supports a wide range of often unique flora. Coastal and inland wetlands cover about 3% of global surface area but provide annual ecosystem services with a value of 47 trillion international dollars (Davidson *et al* 2019), representing 43% of the value of all global biomes. Their immense value stems from their ability to clean water, reduce flood risk, sustain biodiversity hotspots, and

to provide places of natural beauty and wonder that support a range of cultural benefits.

Wetlands also play an important role in the global carbon cycle in terms of net emissions of carbon dioxide and methane, two key greenhouse gases that are central to our goal of limiting global mean increase of 2 °C relative to pre-industrial temperatures. Broadly, the carbon balance of a wetland is associated with changes in vegetation cover, sunlight, temperature, and water coverage (Whiting and Chanton 2001). Emergent vegetation will diurnally and seasonally photosynthetically absorb and respire

CO₂. In open water, the carbon balance is determined by algae, cyanobacteria, aquatic plants, and organic carbon such that when the water concentration of CO₂ exceeds the overlying atmospheric concentration there is a net emission to the atmosphere. Net emission of methane reflects a balance between anaerobic production by archaea (methanogens), as a final stage of the decomposition of organic matter from dead plant biomass, and aerobic and anaerobic loss of methane by bacteria and archaea (methanotrophs) (Guerrero-Cruz *et al* 2021). Transport of methane to the atmosphere is via vascular plants through aerenchyma (e.g. phragmites, papyrus, rice), sediment ebullition, and diffusive fluxes at the water–atmosphere interface, although further work is needed to understand their relative importance on a regional scale. Analysis of field studies suggest that plant-mediated emissions of methane play a key role over tropical wetlands (Helfter *et al* 2022, Shaw *et al* 2022).

Recent studies have highlighted the growing role of tropical wetlands in the global atmospheric growth of methane (Lunt *et al* 2019, 2021, Feng *et al* 2022a), with unprecedented growth rates in 2020 and 2021 being linked with increased emissions from tropical wetland regions (Feng *et al* 2023). The Sudd wetland in South Sudan and the broader Nile Basin, have recently played a disproportionately large role in global methane emissions (Lunt *et al* 2019, 2021, Pandey *et al* 2021), with methane emission during the October–December (OND) short rains season representing, for example, ~30% of the global atmospheric growth rate in 2019 (Lunt *et al* 2021). These changes in emissions are linked with variations in Eastern African rainfall distributions, driven by large-scale changes in sea-surface temperature patterns over the Indian Ocean (Palmer *et al* 2023, and reference therein). A large gap in our understanding of the biogeochemical system is being able to determine changes in the areal extent and vegetative composition of wetlands.

Maps of wetland extent have underpinned spatial emission inventories for methane (e.g. Bloom *et al* 2017). However, these maps tend to be temporally static so they do not capture seasonal or interannual variations in inundation, resulting in errors in methane emissions estimates (Parker *et al* 2020). Satellite Earth Observation (EO) mapping approaches, applied to a time-series of imagery, are able to capture wetland inundation dynamics. One of the most widely used examples is the Global Surface Water dataset (Pekel *et al* 2016), generated using Landsat imagery at a spatial resolution of 30 m giving long-term (since 1984) estimates of surface water extent. However, this approach is limited to the detection of open water. This represents an important shortcoming for mapping wetlands over the tropics where a significant fraction of surface water has

emergent or floating vegetation (Hardy *et al* 2019). Atmospheric measurements have shown that elevated methane emissions from tropical wetlands are highly correlated with these vegetation types (Helfter *et al* 2022, Shaw *et al* 2022).

Satellite EO approaches, such as those based on thermal imaging (Muro *et al* 2018) or signals from global navigation systems (Gerlein-Safdi *et al* 2021), can map both open water and inundated vegetation. But, at spatial resolutions greater than 1 km, these data are unable to resolve hydrological features that underpin wetland system dynamics (di Vittorio and Georgakakos 2021). Inundated herbaceous vegetation can also be mapped using satellite radar imagery at much finer resolutions, typically less than 50 m (Tsyganskaya *et al* 2018, Hardy *et al* 2020, Oakes *et al* 2023). However, satellite radar imaging archives are typically shorter than 10 years in length so that inundation dynamics can only be characterised over relatively short periods. The ability to characterise wetland dynamics over longer periods is crucial for assessing changes caused by climatic fluctuations, or changes in land-use and water management.

Landsat optical imagery can map both open water and inundated vegetation targets at a spatial resolution of 30 m (Díaz-Delgado *et al* 2016, Hardy *et al* 2020). Landsat's near infrared and shortwave infrared channels provide strong signals over water bodies and vegetation with high leaf water content. Landsat acquires imagery for any point on the globe every 16 d but, unlike radar imaging systems, it is affected by cloud cover. As such, Landsat-based mapping approaches are often applied to composite images over a given period (i.e. three month period). However, the multi-decadal and high spatial resolution record of Landsat imagery offers a unique opportunity to reconstruct flood history, providing an unprecedented insight into interannual patterns of inundation and the relative importance of aquatic and inundated vegetation. By using freely available cloud-based services like Google Earth Engine (GEE), this type of analysis can be conducted over large areas (regional or continental). As such, there is a clear opportunity to employ the Landsat archive for understanding how wetlands are contributing to global methane emissions.

Here, we showcase the capability of TropWet, a GEE based approach that harnesses Landsat imaging to deliver serial maps of wetland extent at a spatial scale of 30 m. Our focus is on the temporal and spatial dynamics of the South Sudanese Sudd and its tributaries, covering an area of 242 000 km², that have played a substantial role in the global atmospheric growth rate of methane over the past decade, building on the results of shorter-period studies (Rebelo *et al* 2012, Wilusz *et al* 2017). We also discuss the implications of our results to improving knowledge of the wetland emissions of methane over the region.

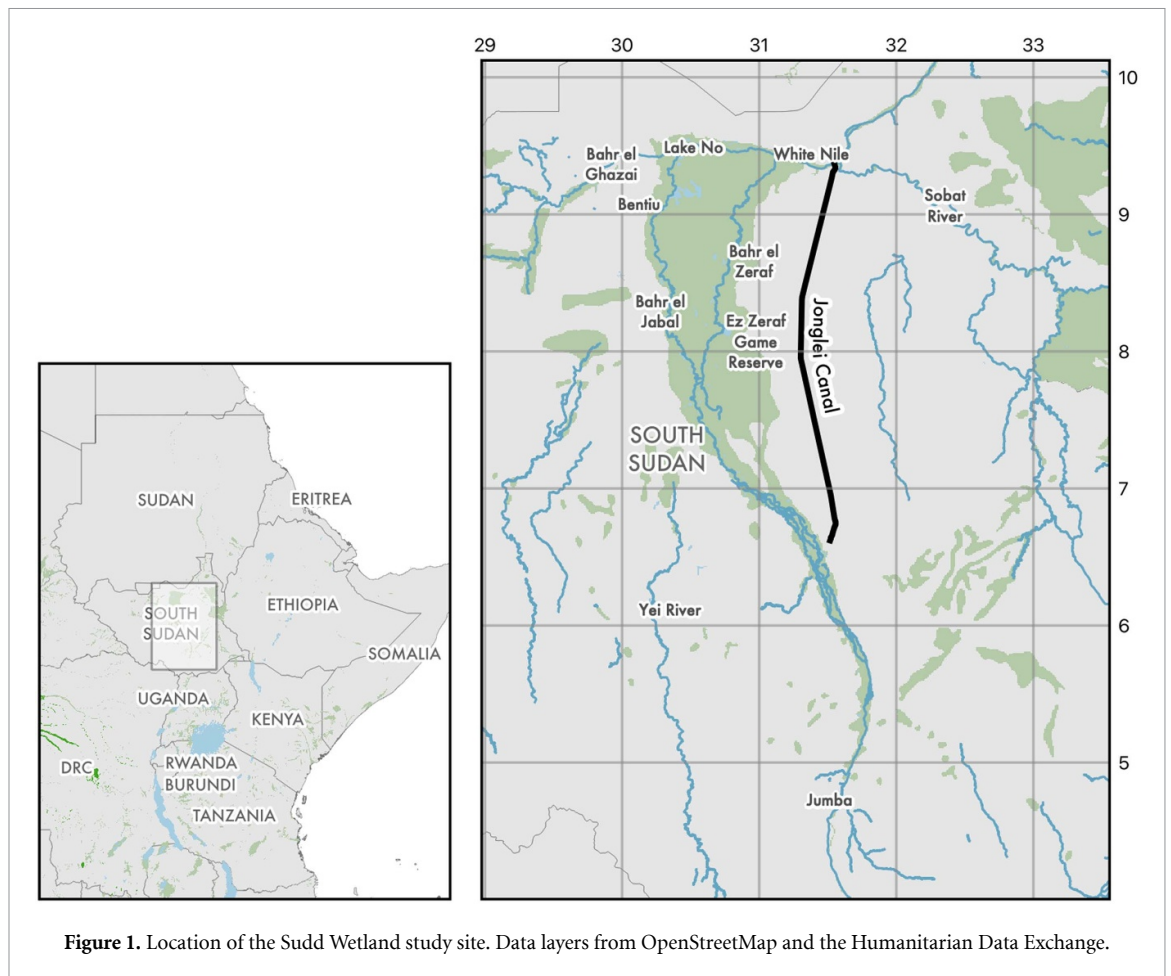


Figure 1. Location of the Sudd Wetland study site. Data layers from OpenStreetMap and the Humanitarian Data Exchange.

2. Data and methods

Here, we describe the Sudd study site and the data and methods we use to study the changes in the extent and composition of the wetlands.

2.1. Study site description

The Sudd floodplain is located on the Bahr el Jebel, the ~1000 km section of the White Nile between Nimule near the Ugandan border and Malakal in South Sudan (figure 1). It is fed primarily by the White Nile, originating from the water inflows to Lake Victoria, one of the two tributaries of the Nile River. Since 1964, a numerical relationship between lake level and outflow under natural conditions (known as the Agreed Curve) has been used to release water from Lake Victoria based on a river gauge in Jinja (Sutcliffe and Petersen 2007); construction of a second dam, due to increasing power demands, led by a modification of the Agreed Curve in 2000 so it released more water (Sutcliffe and Petersen 2007). Departures from this numerical relationship, inferred from satellite altimeter measurements of the lake level, can occur during periods of anomalous high or low rainfall (Vanderkelen *et al* 2018). As the White Nile approaches South Sudan, the water levels overcome the comparatively shallow riverbanks and flow into the shallow and flat inland delta (Sutcliffe and Brown

2018). The Sudd derives its name from the Arabic meaning ‘barrier’, reflecting the low slope angle of the region and the high density of papyrus that effectively forms a barrier to water upstream. Evaporative water loss, inferred from the subtraction of monthly inflows and outflows (Sutcliffe and Parks 1999, Sutcliffe and Brown 2018), can be as large as 50%, consequently greatly reducing upstream water flows in Sudan and Egypt.

2.2. Tropical wetland (TropWet) mapping tool

We use a modified version of the freely available Tropical Wetland mapping tool (TropWet, <https://edin.ac/3RS86KC>) to calculate estimates of wetland extent. TropWet was developed for the geospatial cloud-based platform GEE enabling wetland maps to be produced over large areas (Hardy *et al* 2020). TropWet mainly uses serial imagery from the Landsat series of EO satellites that acquires optical imagery in the visible, near infrared and shortwave regions of the electromagnetic spectrum at a spatial resolution of 30 m. The Landsat archive extends back to 1982 with images collected every 16 d and represents a significant resource for mapping wetlands over broad spatial scales and long time periods.

TropWet is an automated mapping system that applies linear spectral unmixing to spectral endmembers automatically extracted from composite Landsat

Table 1. Description of the TropWet thematic classes relating to inundation extent.

Thematic class	Description
Open water	Surface water, absent of vegetation
Inundated vegetation	Vegetation, particularly herbaceous, emerged in water
Aquatic vegetation	Typically floating vegetation or vegetation canopies completely obscuring the water below
Wet soil	Saturated soil that is unvegetated

satellite optical imagery to determine estimates of fractional cover of water, vegetation and bare soil. A set of logical rules are then applied to the fractional cover product, alongside hydrological terrain metrics and spectral band indices to map inundation extent, including the thematic classes open water, inundated vegetation, wet soil and aquatic vegetation (a description of these classes can be found in table 1). A full description of TropWet can be found in the supplementary materials.

For the purposes of this study, TropWet was applied to the Sudd Wetland using image composites for the months OND, which represent the period of peak inundation for the area (Rebelo *et al* 2012). Annual maps of inundation extent at a spatial resolution of 30 m were generated for the period 1984–2022 except for the years 1985–1987, 1989 and 1994 when there was insufficient cloud-free imagery to form a complete composite.

2.3. Comparison with existing datasets

Most existing datasets that describe wetland extent are temporally static in nature. Here, we compare TropWet-derived maps of wetland extent with commonly used datasets.

GlobCover is a global land cover map generated using 300 m MERIS imagery for 2009, including classes that describe closed/open wetland vegetation and surface water (Arino *et al* 2012). The Global Lakes and Wetlands Database (GLWD) was generated using a combination of sources to delineate lakes, rivers, floodplains and flooded forests (Lehner and Döll 2004). Both the GLWD and GlobCover have been used for modelling global methane emissions (Parker *et al* 2020).

Additionally, Tootchi *et al* (2019) developed static composite wetland (CW) maps for (i) regularly flooded wetlands (RFWs) and (ii) ground-water driven wetlands (GDWs) using a range of existing datasets and hydrological terrain metrics. Below we compare TropWet and the combined RFW-GDW map.

There is a general lack of global datasets that map wetlands with a temporal component. Pekel *et al* (2016) developed the Landsat-based 30 m map of global surface water and how this has changed since 1984. This highly cited work includes a range of metrics including % surface water occurrence. To enable a comparison with this dataset, we generate a % inundation layer for the Sudd using TropWet.

Gerlein-Safdi *et al* (2021) used the L-band signal from Cyclone Global Navigation Satellite System (CYGNSS) satellite constellation to provide monthly inundation maps at a resolution of 0.01 degree (~ 1111 m). Currently, the CYGNSS-based system represents a tractable solution to monitoring wetland extent and has been used to generate improved estimates of methane emissions over the Sudd (Gerlein-Safdi *et al* 2021). We compare CYGNSS maps of wetland extent with those generated by TropWet.

The Gravity Recovery and Climate Experiment (GRACE) mission provides an estimate of water storage at ~ 30 d intervals at an approximate spatial resolution of ~ 300 km (Wahr *et al* 1998, Syed *et al* 2008). Whilst the spatial granularity of the GRACE water thickness product is much coarser than TropWet, we compare the two outputs from a temporal perspective. We also conducted a temporal comparison between TropWet inundation extent and altimetry based lake height at Lake Victoria generated through the Global Reservoirs and Lakes Monitor (G-REALM) program that reconstructs lake height using a range of satellite altimetry systems (Duan and Bastiaanssen 2013).

3. Results

We report year to year changes in inundation extent from 1984 to 2022 over our study region that is centred on the Sudd wetland (figure 1), including changes in vegetation cover. We evaluate our wetland extent data product using other commonly-used data products.

3.1. Regional estimates of inundation extent

Yearly maps of inundation extent (see ‘supplementary results document’ for individual maps) for the OND short rains period demonstrate considerable variation over the period 1984–2022, including a dramatic increase in extent since 2019 (figure 2). The year-to-year variability in extent is consistent with estimates of water thickness made by the GRACE mission between 2002 and 2021 (figure 3) (Pearson correlation = 0.92, $p < 0.01$). Both TropWet and GRACE indicate a trend in increasing inundation extent.

The largest flooding events observed by TropWet all occurred within the period 2019–2022, representing 233%–321% increase from the pre-2019 OND median inundation extent. Since 2009, the observed trend of increasing inundation coincides

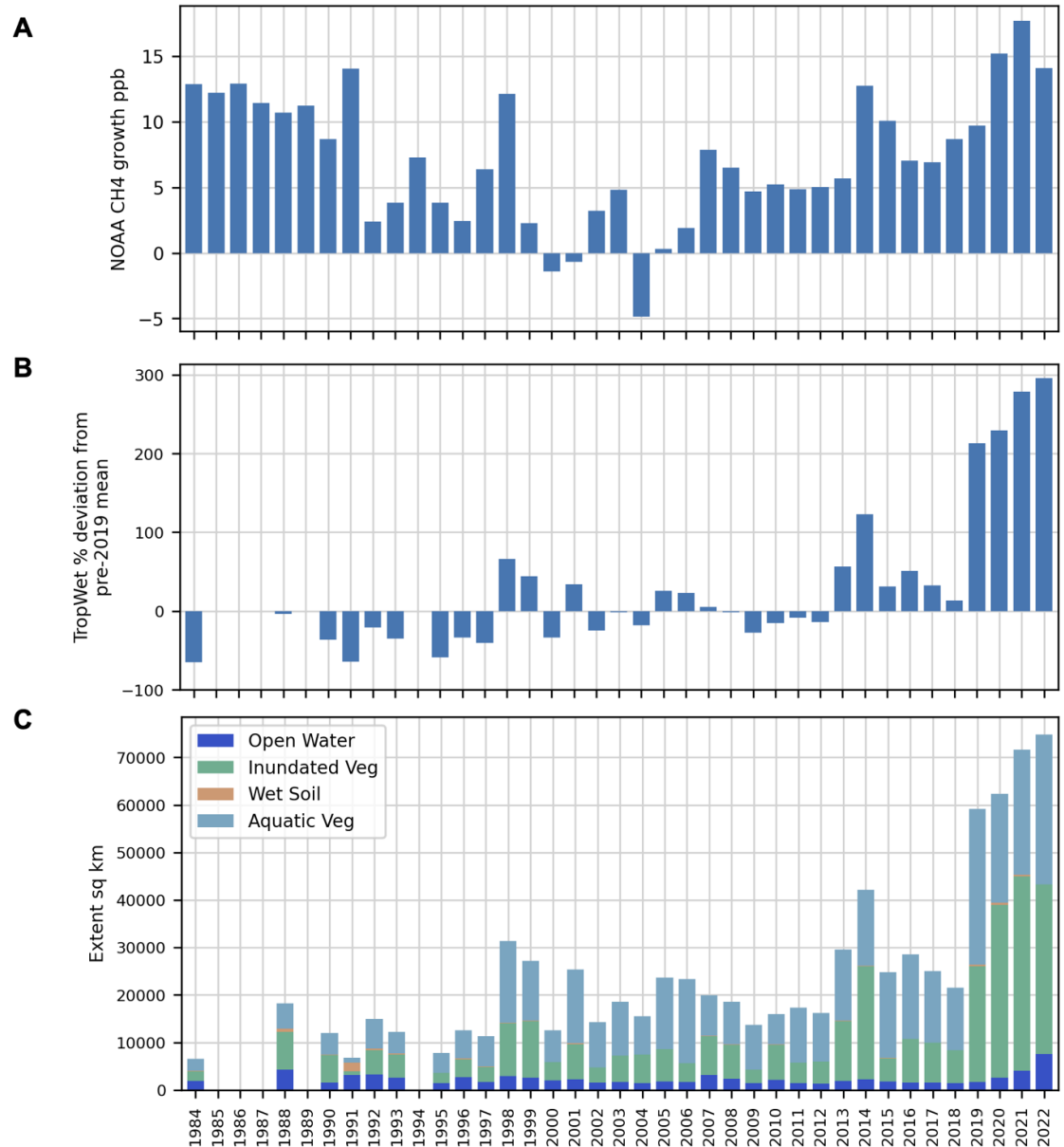


Figure 2. Plots demonstrating (A) NOAA annual growth rate in global methane (ppb), (B) TropWet OND inundation extent, expressed as a % deviation from the 1984–2022 mean extent and (C) stacked bar chart showing the overall inundation extent per year (OND), summarised by inundation type: open water, inundated vegetation, wet/saturated soils and aquatic vegetation.

with a recent general increasing trend in global methane emissions observed by the US National Oceanic and Atmospheric Administration (NOAA) (figure 2) (Pearson correlation 0.88, $p < 0.01$) although little or no trend is observed before 2009.

We find that TropWet inundation extent estimates for OND are correlated with mean quarterly values of Lake Victoria lake levels since 1992 (figure 4). A lagged correlation analysis resulted in Pearson correlation values ranging from 0.69 to 0.73 (lags of 0 to -3 quarters), with the largest correlation of 0.73 ($p < 0.001$) with no lag. The well-documented El Niño event of 1998 led to a marked increase in Lake Victoria levels due to an increase in rainfall within the Lake Victoria basin (Awange *et al* 2008) which is mirrored in the TropWet flood event recorded for

that year. More recently, the period 2019–20 has seen the highest levels recorded on Lake Victoria due to increased rainfall in response to successive positive Indian Ocean Dipole (IOD) phases (Khaki and Awange 2021). Again, this pattern is reflected in TropWet estimates of inundation extent for the Sudd.

3.2. Seasonal maps of inundation extent estimates

Using the annual thematic classification maps of inundation extent (see the supplementary results document for individual maps) the mode class was extracted over the 1984–2022 time series to provide a map of inundation extent for a ‘typical’ OND short rains season. Image differencing was then used to determine spatial trends of annual flooding that deviates from the mode. Figure 5 shows the

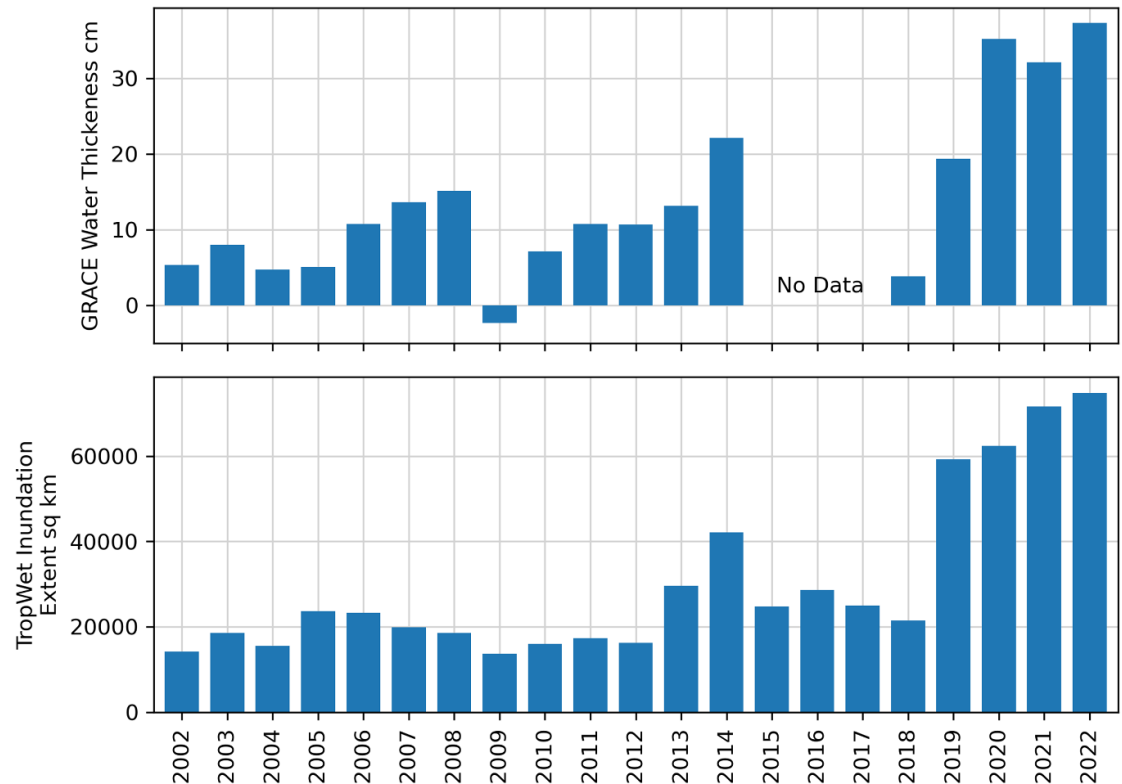


Figure 3. Bar charts comparing mean OND GRACE water thickness (cm) (7 N 30E–8 N 31E) against TropWet inundation extent (km^2).

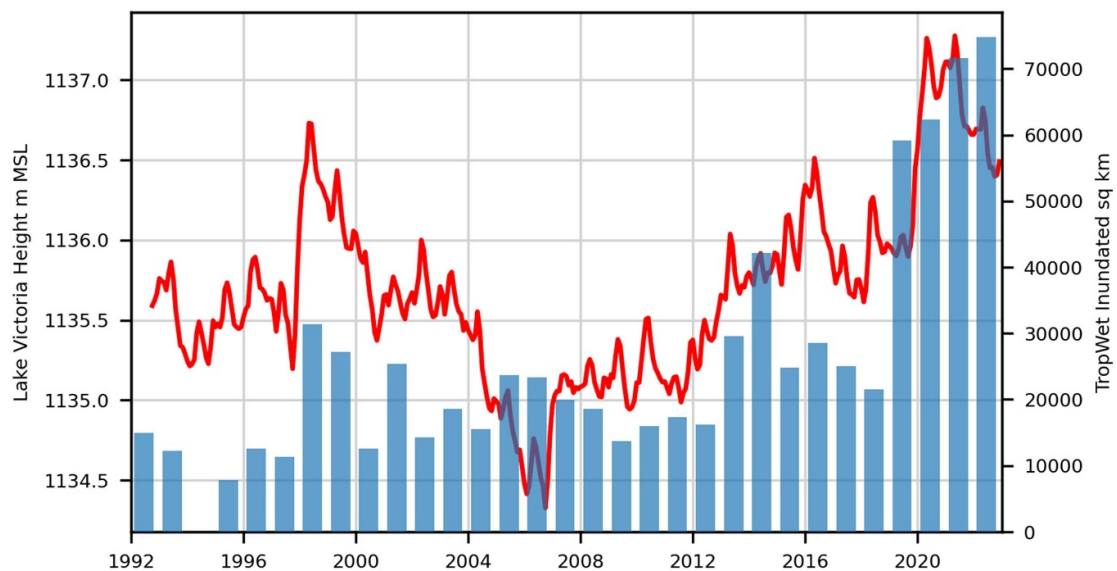


Figure 4. A comparison between monthly mean values for the height of Lake Victoria (m MSL, red line) against OND values for TropWet inundation extent estimates (km^2 , blue bar graph).

map for 2022 corresponding to the largest inundation extent ($74\,820\text{ km}^2$, figure 3) recorded since 1984. Much of these changes occur in two regions: (i) the area between the Bahr el Zeraf channel and the Jonglei Canal, a result of overbank flooding from the main Bahr el Zeraf channel; and (ii) the area towards the west of the study location between the Bahr el Jabal and Bahr el Ghazai channels, possibly resulting from

backflow where runoff reaches Lake No and the Bahr al Abyad section of the White Nile.

A significant proportion ($22\,770\text{ km}^2$) of the recent increase in inundation (compared to the 1984–2022 mode) occurs outside of the floodplain region defined by static datasets like the GLWD (figure 6). Additionally, we find that recent changes in inundation extent are dominated by inundated vegetation

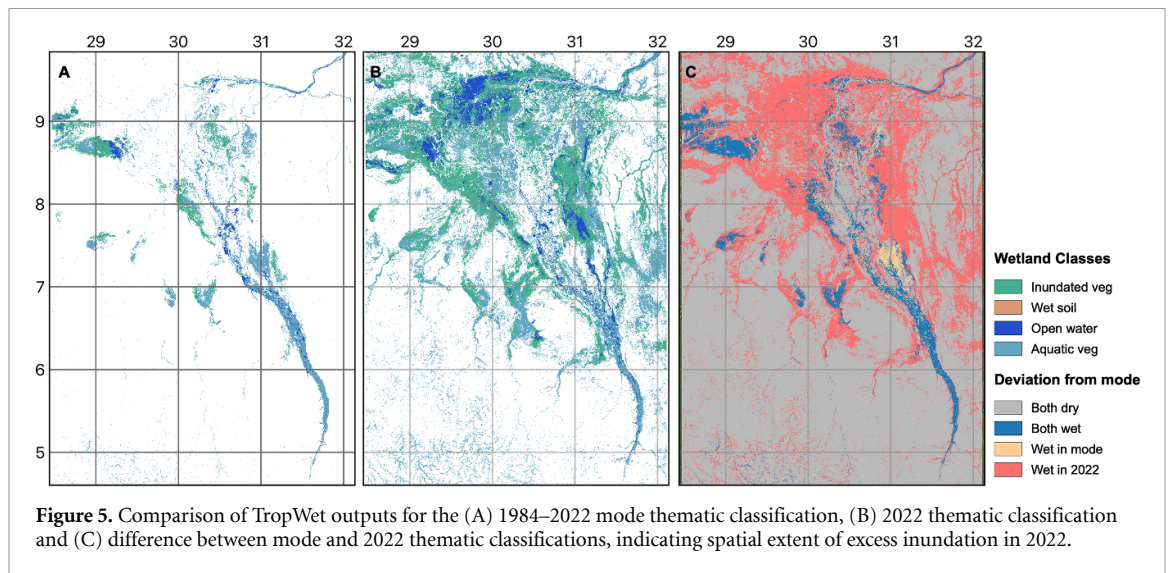


Figure 5. Comparison of TropWet outputs for the (A) 1984–2022 mode thematic classification, (B) 2022 thematic classification and (C) difference between mode and 2022 thematic classifications, indicating spatial extent of excess inundation in 2022.

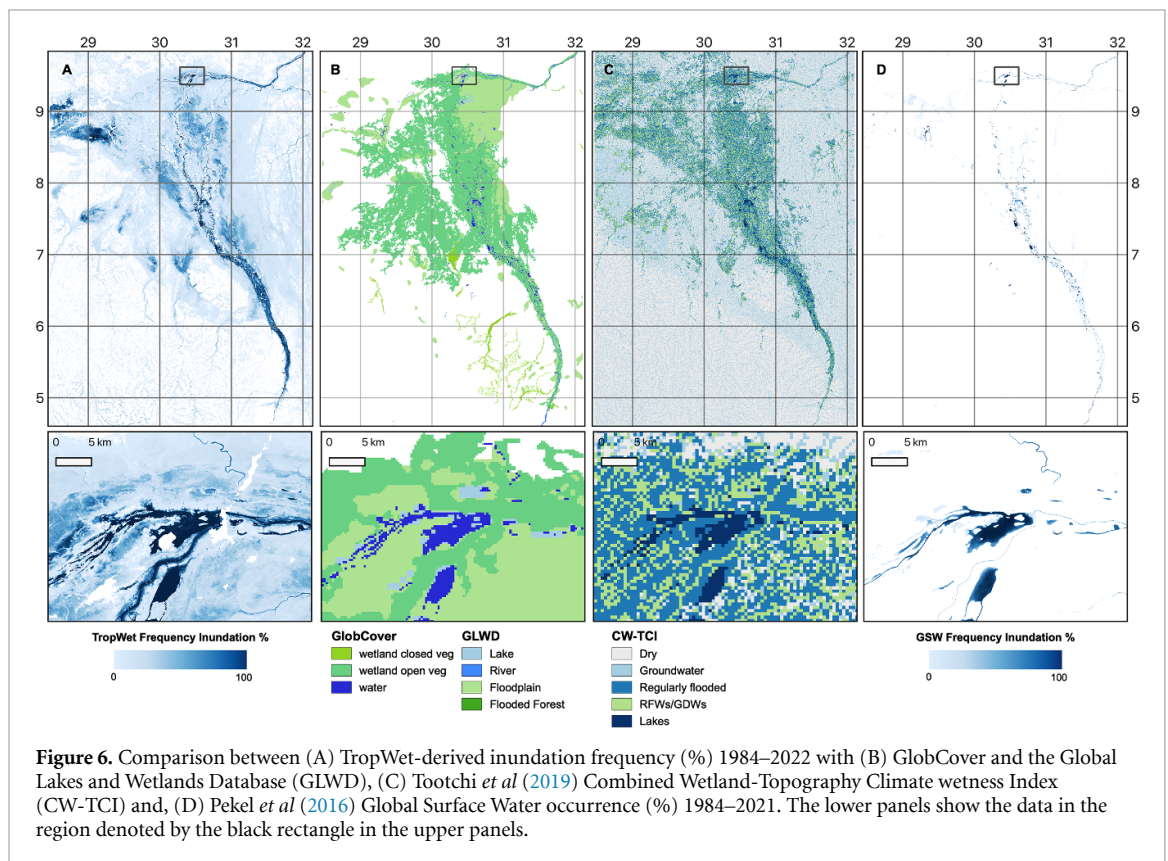


Figure 6. Comparison between (A) TropWet-derived inundation frequency (%) 1984–2022 with (B) GlobCover and the Global Lakes and Wetlands Database (GLWD), (C) Tootchi *et al* (2019) Combined Wetland-Topography Climate wetness Index (CW-TCI) and, (D) Pekel *et al* (2016) Global Surface Water occurrence (%) 1984–2021. The lower panels show the data in the region denoted by the black rectangle in the upper panels.

and aquatic vegetation (figure 2 and the supplementary results document), accounting for an average of 90% of total inundation (40%: inundated vegetation, 50%: aquatic vegetation). This trend is particularly apparent in the floodplain regions, away from the main trunk river channels.

3.3. Comparison against existing products

Figure 6 provides a comparison between TropWet-derived inundation frequency for OND for the period 1984–2022 against GlobCover (Arino *et al* 2012), GLWD (Lehner and Döll 2004), CW-TCI (Tootchi *et al* 2019) and GSW (Pekel *et al* 2016). Whilst

the static inundation maps demonstrate a visually broad agreement with TropWet, these products do not describe the temporal variability in inundation that are experienced by wetlands like the Sudd.

TropWet inundation extent estimates are comparable with those from CW-TCI (500 m) (table 2), but TropWet is able to provide this information at a 30 m spatial scale that is necessary to provide insights into the hydrological drivers behind inundation extent and its temporal dynamics. For example, TropWet is able to demonstrate the hydrological influence of the Jonglei Canal embankment that spans the Sudd as well as fine scale anabranching features (figure 7).

Table 2. Comparison of inundation extent estimated by TropWet and a range of other products: Tootchi *et al* (2019) Combined Wetland-Topography Climate wetness Index (CW-TCI), Pekel *et al* (2016) Global Surface Water (GSW) occurrence, GlobCover and the Global Lakes and Wetlands Database (GLWD).

Product	Extent km ²
TropWet OND Frequency Inundation >1%	127 673
TropWet OND Frequency Inundation >5%	94 504
TropWet max extent 2019–2022	74 820
TropWet max extent 1984–2018	42 128
CW-TCI	110 203
CW-TCI non-groundwater	48 141
GlobCover	42 803
Global Lakes and Wetlands Database (GLWD)	31 552
GSW occurrence >1%	473

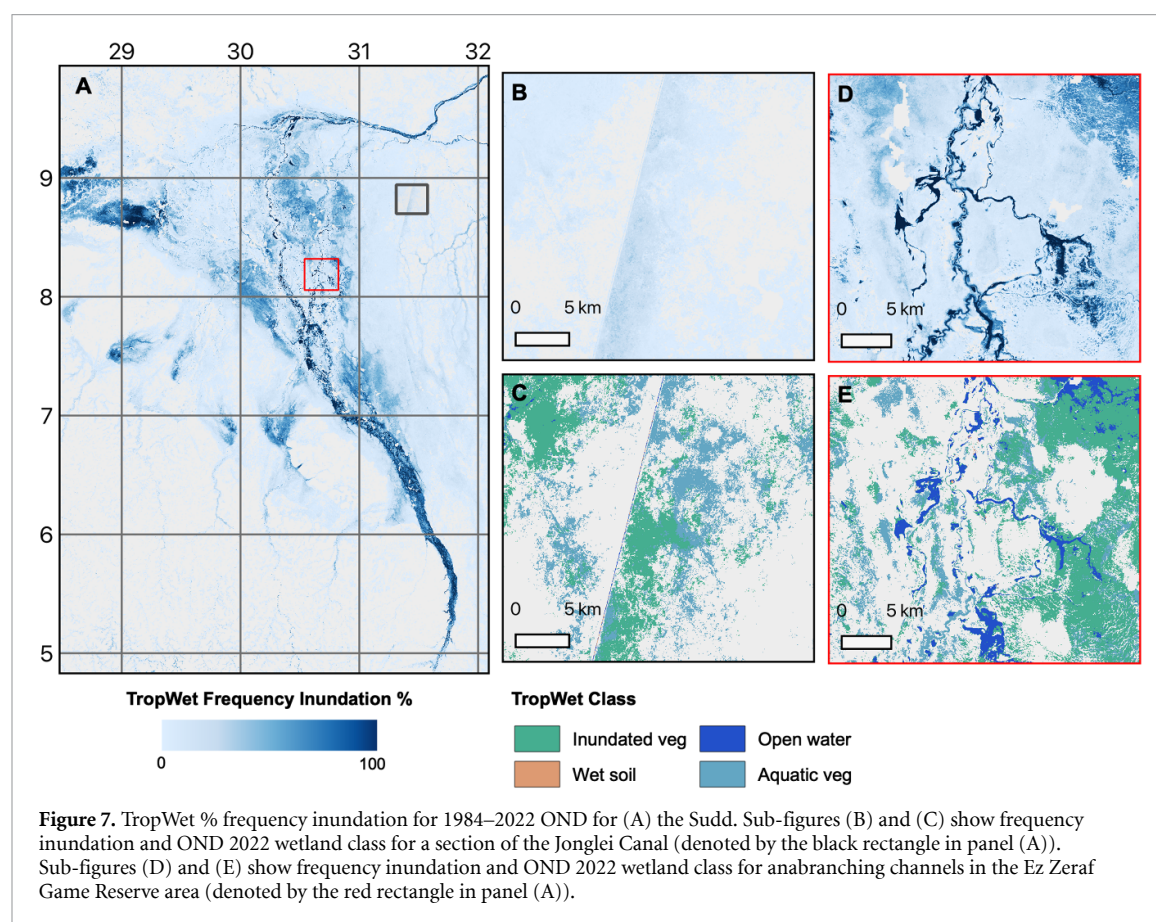
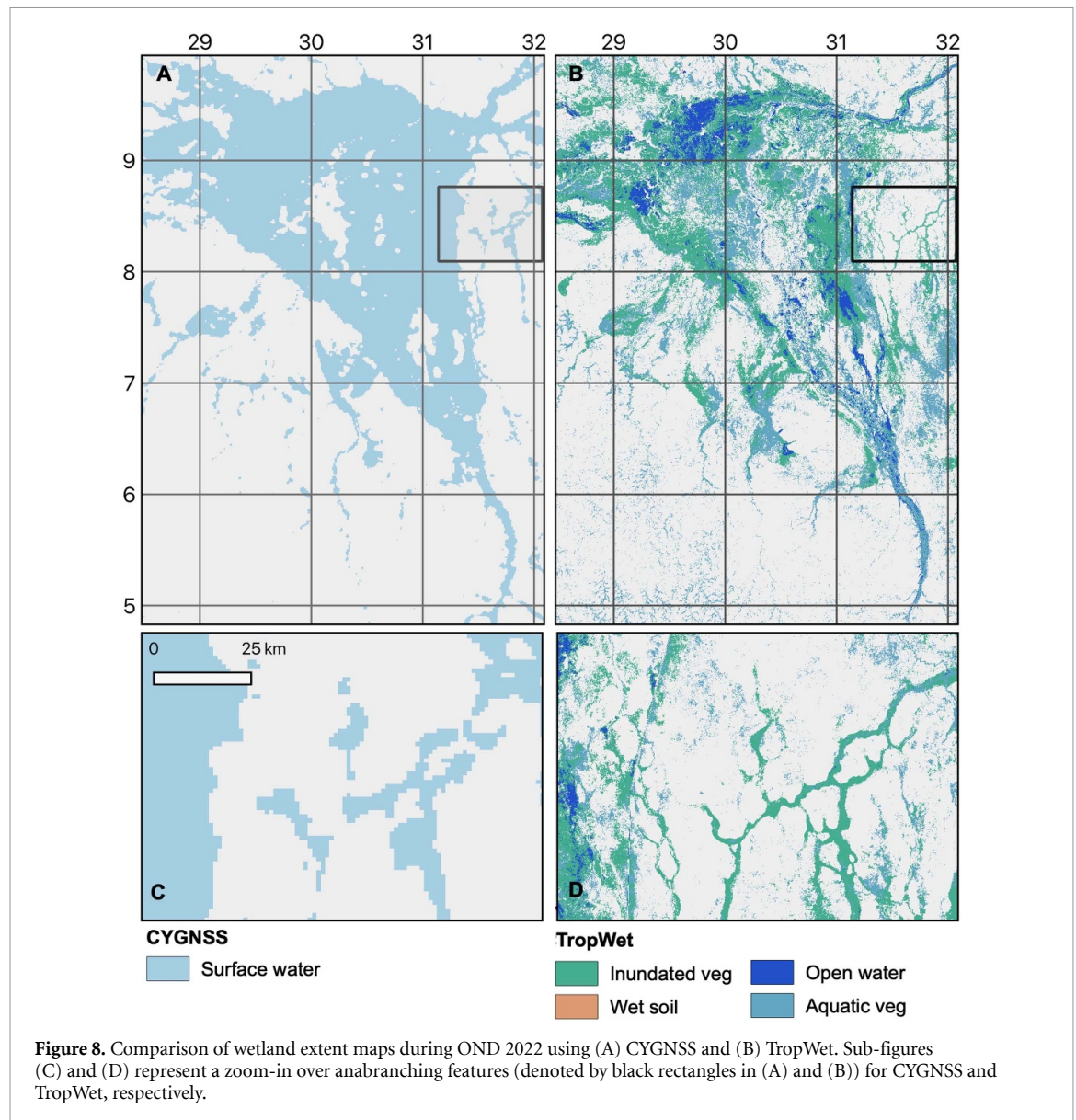


Table 3. Comparison of CGNSS and TropWet maximum wetland extents for the OND period between 2018 and 2022.

Year OND	CYGNSS extent km ²	TropWet extent km ²	% difference
2018	59 793	21 502	64%
2019	55 626	59 204	−6%
2020	73 368	62 347	15%
2021	78 042	71 589	8%
2022	76 876	74 820	3%

We find that the TropWet frequency inundation is much higher than GSW occurrence layer, with GSW reporting an extent of only 473 km² (table 2). This is due to GSW being limited to open water detection, thereby underlining the importance of also mapping inundated vegetation and aquatic vegetation in wetland inventories.

We find good agreement between inundation extent estimates from TropWet and CYGNSS (Gerlein-Safdi *et al* 2021), 2019–2022, with differences typically within 10% (table 3). In 2018 CYGNSS estimates 64% more inundation than TropWet. We do not currently understand the reason for this discrepancy but the smaller TropWet extent for 2018 is more



consistent with changes in the GRACE water thickness metric. The CYGNSS signal is likely to respond to saturated soils, as well as surface water (Gerlein-Safdi *et al* 2021), unlike TropWet that only responds to surface inundation. The CYGNSS approach, using the L-band microwave signal, operates independently from cloud conditions and therefore achieves high temporal granularity (monthly), representing an improvement over the optical satellite based TropWet approach that used composites made over a three-month period. However, the CYGNSS outputs are generated at a comparatively coarse spatial resolution (~ 1111 m compared to 30 m for TropWet) meaning that fine scale hydrological features, including major tributaries or anabranches, may not be captured (figure 8). Importantly, for applications such as estimating methane emissions, the CYGNSS approach cannot distinguish between open water and inundated vegetation.

4. Discussion and conclusions

We demonstrated the use of TropWet, a freely available EO wetland mapping hosted on the cloud-based platform GEE, for mapping fractional cover of water and vegetation in East Africa. Employing the Landsat archive of optical satellite imagery, TropWet offers key benefits over existing approaches in terms of its high spatial resolution (30 m), the ability to discriminate inundated vegetation from open water and the ability to map wetland extent over long periods at three-monthly time-steps. As such, TropWet represents a scalable solution to fine-scale mapping of wetlands dynamics, representing a significant tool for a number of applications including estimates of global methane emissions.

The focus of our analysis is mapping the Sudd wetland in South Sudan that has been linked to disproportionately large seasonal methane emissions

(Lunt et al 2019, 2021, Pandey et al 2021, Feng et al 2023, Palmer et al 2023). The Sudd wetland is fed by seasonal rainfall, particularly in the second half of the calendar year, and by inflow from the White Nile (Bahr el Jebel) (Rebelo et al 2012). Variation in the flood pulse has led to significant changes in the coverage of inundated vegetation, which is now recognized as a key factor behind the magnitude and spatial distribution of methane emissions from natural wetlands (Helfter et al 2022, Shaw et al 2022).

We used TropWet to reconstruct the OND flood history from 1984 to 2022 over the Sudd demonstrating excellent agreement with regional estimates of water storage made by the GRACE mission. Our analysis revealed considerable inter-annual variability in the inundation extent of the Sudd wetland, ranging from 6570 km² in 1984 to a peak of 75 000 km² in 2022, with a standard deviation of 17 600 km² over our 38 yr study period. Our findings have profound implications for methane modelling that typically relies on static wetland data products such as GlobCover and the GLWD and therefore could lead to biases in wetland emission estimates (Bloom et al 2017, Parker et al 2020). The year-to-year variation in wetland extent of the Sudd also bears a striking resemblance to the record of global mean atmospheric growth of methane since the turn of the century, supporting previous analysis of satellite observations of methane since 2010 (Lunt et al 2019, Feng et al 2022a, 2022b, 2023). Previous work estimates that methane emissions for South Sudan, which is dominated by the Sudd, contribute to 13%–14% (3.4 Tg methane/year) of the total emissions for East Africa (25–27 Tg methane/year) between 2018 and 19 (Lunt et al 2021). Future work should reconstruct flood histories for other wetland systems in the East Africa region to understand their collective role in driving the recent increase in methane emissions.

We found that since 2019 the increase in flooded extent is dominated (~90% of newly flooded areas) by inundated vegetation and aquatic vegetation, which are land cover types associated with the net production of methane (Helfter et al 2022, Shaw et al 2022). This result underlines an important shortcoming in current attempts to model global methane emissions that typically rely solely on maps of open water extent. Failure to account for the role of inundated vegetation increases the probability of erroneous emission estimates. Existing approaches for mapping the temporal variability in wetland extent, notably including the CYGNSS-based approach (Gerlein-Safdi et al 2021), do not discriminate between open water and inundated vegetation. TropWet is able to provide information about inundated vegetation cover and therefore has potential to significantly improve models of wetland methane emission. Confidence in current TropWet outputs is based on validation sites elsewhere in

Africa (Zambia and Tanzania: Hardy et al 2020) and through corroboration with existing temporal products (i.e. GRACE Water Thickness and Lake Victoria height data). We acknowledge that uncertainty will exist in the inundation extent maps for the Sudd due to the absence of regionally-specific field data. Future research should focus on evaluating TropWet using field-based measurements of inundated vegetation from regions like South Sudan and on how subsequent TropWet derived inundation maps can improve broadscale methane models.

Previous work has identified the Sudd grassland floodplains to be dominated by macrophytes, including *Oryza longistaminata*, *Echinochloa pyramidalis* and *Hyparrhenia rufa*, and more permanently inundated areas to be dominated by *Cyperus papyrus*, *Phragmites communis*, *Vossia cuspidata*, *Typha domingensis* and *Eichornia crassipes* (Rebelo et al 2012). Many of these species are known to provide efficient pathways for methane into the atmosphere (Koné and Borges 2017, Helfter et al 2022). Additionally, species such as *Typha* (cat tail or bulrush) and *Eichornia crassipes* (water hyacinth) are invasive and their spread has had significant impacts on African wetlands (Struik et al 2022). The relative importance of individual types of vegetation in driving methane emissions has yet to be established. By virtue of its spatial resolution and ability to discriminate aquatic vegetation and inundated vegetation, TropWet can be used to develop new insights into landscape stratification and to help direct ground-based field campaigns on the species dominating newly flooded areas in the Sudd and how they influence landscape scale methane emissions.

In this study we focused on the short rains season (OND) that corresponds to peak methane emissions (Lunt et al 2019, 2021). We applied TropWet to Landsat composites for that three-month period. Hydrological dynamics occur at much finer timescales for most floodplain systems. While Landsat imagery is acquired every 16 d, changes in cloud cover preclude our ability to routinely report composite images on timescales shorter than three months; although in some years, relatively low cloud coverage may permit TropWet analysis at a finer timescale offering new insights into mechanisms behind flood events. Combining Landsat imagery with other sources of optical imagery, particularly Sentinel-2, would increase the rate of image acquisition to approximately three days, thereby increasing the chance of generating a cloud-free composite over periods shorter than three months. Generally, optical imagery is limited by cloud cover. As such, radar imaging systems, which are unaffected by cloud cover, represent important sources of information for mapping wetlands at a high temporal and spatial resolution (Hardy et al 2019, Oakes et al 2023) despite the relatively short length of archive. A hybrid approach that combines optical and radar

sources represents a promising direction for future research, particularly as routine acquisition of L-band imagery becomes available (a joint NASA-Indian Space Research Organisation system is due to be launched in January 2024), giving greater canopy penetration and estimates of inundated vegetation in complex landscapes.

In East Africa, extreme flooding events are linked to the positive phase of the IOD, which is expected to be more frequent and more extreme with climate change (Cai *et al* 2014, Palmer *et al* 2023). This will have significant consequences for methane emissions from natural wetlands, as well as severe implications for other factors such as disease (particularly mosquito-borne diseases like Malaria), loss of pasture for livestock, disruption of critical infrastructure and the displacement of people (Patz and Olson 2006, Wainwright *et al* 2021). We have demonstrated TropWet's ability to quantify long term changes in inundation dynamics for large floodplain environments like the Sudd. In doing so, TropWet can have a key role in understanding how wetlands are responding to our changing climate and, in doing so, help to inform and design mitigation strategies and improve estimates of greenhouse gas emissions.

Broadly speaking, our analysis has highlighted the added value of the Landsat EO imagery, interpreted using the TropWet tool, to improve understanding of hydrology that contributes to seasonal and year-to-year changes in wetlands over a region where we have little additional information. The imagery data provide new information to interpret contemporary changes in the atmospheric growth rate—in our example, year-to-year changes in the areal extent of the Sudd and surrounding wetlands are remarkably consistent with observed changes in the global atmospheric methane growth rate. This supports the idea that a large fraction of those recent annual changes in the atmospheric growth of methane are due to wetland emissions driven by changes in hydrology, particularly from Eastern Africa (Lunt *et al* 2021, Pandey *et al* 2021, Peng *et al* 2022, Qu *et al* 2022, Feng *et al* 2022b, 2023). The EO imagery data also represent additional information with which to constrain computational model parameters to improve the predictive capability of those models to describe hydrodynamics in continental-scale rivers. In doing so, they have the potential to improve short-term flood forecasts over the region and also to reduce the impact of future climate on the regional hydrology and subsequently on wetland emissions of methane.

Data availability statement

The data that support the findings of this study are openly available at the following URL/DOI: <https://doi.org/10.20391/ba61e746-d663-4b25-88d1-204dfd051fbd>.

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Ethical statement

This study was approved by the Aberystwyth University Research Ethics Panel (ID: 26125).

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